

CS07340 MID-TERM EXAM FOCUS POINTS

1. Fundamentals of the Analysis of Algorithm Efficiency (Quizzes 1, 2, 4 – 6, Homework 2)

- a. Algorithm determinism
 - What can be calculated with an algorithm?
 - What can be approximated by an algorithm?
- b. Step-counting
 - Given pseudo-code, count instructions.
- c. Asymptotic Big-O Notation
- d. Run-time Complexity Big-O Notation
 - Run-time complexity of standard algorithms.

2. Exhaustive Search and Conquer Algorithms (Quizzes 2 – 5)

- a. Programming Paradigms
- b. Master Theorem
 - Application given a recurrence pattern.

3. Balanced Search Trees and Related Structures (Quizzes 6, 7)

a. Search Tree Properties and Run-time Complexities

- Binary Search Trees
- AVL Trees
- 2-3 Trees
- Red-black Trees